

Orbital Oscillation: Dematerialisation of Craft through Imperceptible Actuation of a Glass Object

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Figure 1: Orbital Oscillation installation showing the glass object on white table top with thin metal legs. The robotic mechanism is hidden in the thickness of the table.

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Abstract

Orbital Oscillation is a kinetic sculpture that explores the phenomenology of time through a deceptively simple act: a glass object rolling in slow, deliberate orbits on a round table. The object's movement is not self-initiated but subtly actuated by a concealed robotic system beneath the table's surface. Using real-time sensing and feedback control, the table imperceptibly tilts to inject energy into the glass object's trajectory. On the one hand, this paper presents the mechatronic and algorithmic design of the system, detailing how sensing, control, and form are orchestrated to create an experience that appears autonomous, fluid, and continuous, yet is

precisely choreographed. On the other hand, the conceptual framework of Digital Craft is discussed, placing the work in the context of perception through Husserl, and a critical engagement with technology through Heidegger and Hui. The work invites reflection on how machine intelligence can shape perception without revealing its own presence.

CCS Concepts

• **Applied computing** → **Media arts**; • **Computer systems organization** → *Robotic control*.

Keywords

kinetic sculpture, Digital Craft, glass art, mechatronic design, feedback control, imperceptible motion, phenomenology

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1 Introduction

Orbital Oscillation (Fig. 1) is a kinetic sculpture that explores motion, perception, and the phenomenology of time through digitally choreographed behavior. A hand-blown glass object—shaped like a hybrid sphere—rolls in slow, deliberate orbits across a round table. Its apparent autonomy masks a concealed mechanism: a low-profile actuation system embedded within the table surface subtly tilts it to speed up or slow down the rotation. The resulting motion appears spontaneous and cyclic, yet is precisely orchestrated. Inspired by Edmund Husserl's notion of internal time consciousness—retention, impression, and protention—the sculpture becomes a meditation on the tension between predictability and disruption, presence and expectation. This paper situates Orbital Oscillation within a broader lineage of the emerging discipline of Digital Craft, particularly as sculptural investigations into temporality, agency, and perception. We describe how the system's material design and control logic work in service of conceptual goals, using embedded sensing, phase-based actuation, and invisible causality to foreground the tensions between determinacy and surprise.

1.1 Artistic Context and Prior Work

Orbital Oscillation extends a long-standing inquiry into the temporal, perceptual, and material behaviors of digitally augmented sculptural objects. Previous works by the artist have investigated how computational systems may be used not merely for control or fabrication, but as instruments for articulating material processes across time. In particular, the Glass Entanglement series explored how 3D scans of cut glass structures allow reconfiguration into new spatial compositions through 3D-printed scaffolds (Fig. 2). The result defined digital craft in which the computer did not generate new geometry, but instead assisted in the assembly and transfer of residual traces from earlier transformations—suggesting that matter itself retains a kind of memory.

Subsequent reverse experiments from the series introduced 3D-printed stainless steel forms into the hot-shop process as lost tools



Figure 2: Glass Entanglement I (2020). Digitally reconstructed glass fragments held apart by a 3D-printed scaffold. The scaffold's geometry is derived from 3D scans of the fragments arranged in virtual space, exploring how digital tools can preserve and recompose material memory.



Figure 3: Prototype of a stainless steel cage, 3D printed and used as a lost tool in the process of glass blowing. This method extends traditional cage-blowing techniques into the realm of digital constraint and thermal deformation.

in the blowing of glass, reinterpreting historical cage-blowing techniques with digitally generated constraints (Fig. 3). In both cases, the temporal quality of the material—the slowness of deformation, the capture of thermal expansion, the freeze of soft form—was central to the expressive logic of the work. These projects resonate with Richard Sennett's call for a return to craftsmanship as a negotiation between resistance and intention [Sennett 2008, p. 11]. In Orbital Oscillation, this focus on temporal expression shifts from shaping material to choreographing behavior. The glass object—now a rolling, closed form—is no longer placed in a visible cage, but instead constrained by its own rolling motion. The emphasis is extended beyond its 3D material form towards a 4D trajectory that unfolds in time—an act we refer to as 'dematerialization of craft'. By hiding the robotic mechanism and modulating motion through

minimal intervention, the work evokes a sense of internal logic, as though the object were self-moved. The choice of clear transparent glass heightens the tension between what is seen and what is caused, as the object reveals nothing within.

1.2 Philosophical Framework: Time, Perception, and Hidden Technics

The perceptual structure of *Orbital Oscillation* is grounded in phenomenological understandings of time and technology. The work draws directly on Edmund Husserl's triadic model of temporal experience—retention, impression, and protention—in which perception is never a static moment, but an unfolding that integrates memory, presence, and expectation [Husserl 4 30, p. 33]. As the glass object completes arcs and reversals, viewers begin to internalize a rhythm in the rolling sphere. They anticipate the return of the object, and when that expectation is disrupted—either delayed, accelerated, or reversed—attention is redirected toward the system as a whole and the perceived danger of its coming out of the system and potentially failing through falling from the table. In this way, the artwork performs a kind of temporal phenomenology, inviting the viewer to sense not only the motion of the object, but the structure of their own anticipation and tension in the potential destruction of it. This engagement with hidden causality aligns with Martin Heidegger's critique of modern technology as *Gestell*—a mode of enframing that reduces the world to resource and function [Heidegger 1977, p. 19]. By contrast, *Orbital Oscillation* resists instrumental transparency. It refuses to explain itself. Instead, it renders technical agency imperceptible: the motors make no sound; the table's tilt is invisible; the object appears autonomous. In doing so, the work revives Heidegger's alternate notion of *techne* as a form of revealing—technology not as tool, but as poetics of disclosure, essential in the description of digital craft [Klein 2018, p.336–342]. Finally, the work can be situated within Yuk Hui's concept of cosmotechnics—a call to reimagine technological systems not as globally standardized infrastructures, but as culturally situated ways of ordering time, matter, and thought [Hui 2021, p. 9]. Rather than foregrounding optimization or interactivity, *Orbital Oscillation* adopts a rhythm of drift and return. Its behaviour emerges not from command, but from coordination; not from precision, but from phase. The system's choreography mirrors not industrial automation, but planetary motion—cyclical, entropic, and minimally insistent.

2 Mechatronic Design Decisions

The sculpture's mechanical design is driven by a core artistic desire: to animate the glass object without any visible moving parts or external intervention. The actuation and control system can be compared to a classical ball-on-plate balancing system that necessitate real-time feedback control [Awtar et al. 2002]. However, the focus of our system are the hidden sensors (pose estimation for the glass object) and actuators (to roll the object). The round table surface serves both functional and perceptual roles—its circular form masks minor tilt adjustments and emphasizes the continuous, orbital motion of the object. On the table, a custom-blown double-sphere glass object naturally rolls along a circular path due to its asymmetric geometry—a form chosen for its stable and repetitive trajectory. This effectively reduces its degrees of freedom to a single angular

dimension around the table's centre, simplifying both sensing and control. The minimalist table legs that resemble the base of Eames' Wire Chair further contribute to the magic by revealing the absence of an external power source.

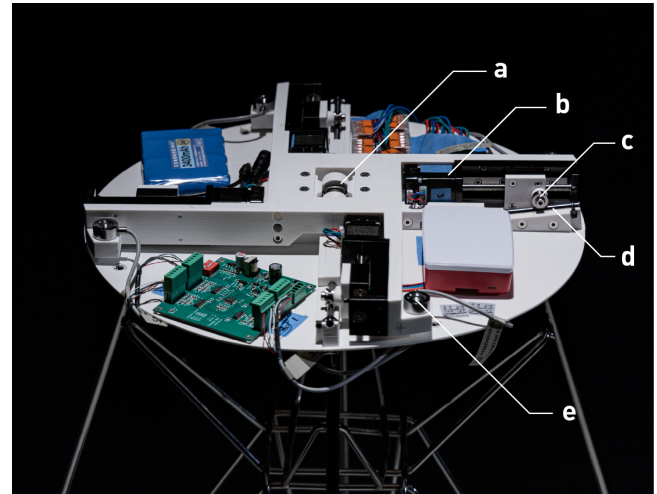


Figure 4: Photo of the robotic tilting mechanism that is hidden within the table thickness. Main components are shown, including (a) universal joint bearing, (b) linear actuator, (c) V-groove bearing and (d) tilted guide rod, and (e) load cells.

To achieve imperceptible, silent actuation, the tabletop is mounted on a custom-designed, low-profile robotic tilting mechanism concealed within a 45mm thick table top (Fig. 4). A universal joint bearing at the centre allows for two degrees of freedom in tilt—azimuth (bearing angle of the low point) and amplitude—while preventing unwanted pan (rotation around Z axis) and translation motion (in X Y Z directions). Four linear actuators are arranged at the cardinal corners of the table. Each actuator is powered by a 28mm-size (NEMA11 equivalent) closed-loop stepper motor with integrated trapezoidal leadscrew. The screw has a 4mm lead and a total travel of 50mm. To minimize noise, the stepper motors are driven using field-oriented control (FOC) to minimize torque ripple, resulting in motion that is inaudible from the viewer's position. Backlash in the lead screws is mitigated using spring-loaded tensioners, further ensuring quiet reversals.

Mounted on each actuator's moving platform is a V-groove bearing that rides on top of a fixed guide rod placed at a 7.7° incline (Fig. 5). This mechanical translation transforms linear motion into vertical displacement, which in turn tilts the tabletop around the central bearing. Each opposing actuator pair (X- and Y-axis) is coordinated such that one side extends while the other retracts, creating a planar tilt in the desired direction. By combining the movements of both actuator pairs, the system can produce smooth tilts in any azimuth at a resolution below 0.05°. In order to allow tilting in both directions, it is essential that the guide rod has a round profile, and that the contact point is allowed to tilt around it. The system's maximum designed tilt is $\pm 1.2^\circ$. During operation, the maximum tilt is capped at $\pm 0.8^\circ$ to maintain perceptual subtlety.

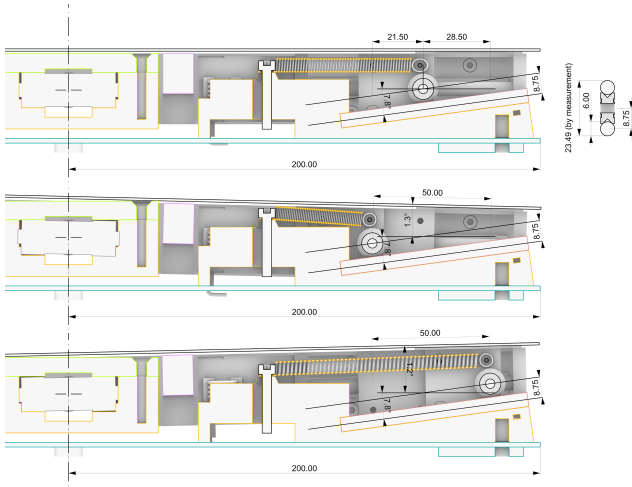


Figure 5: Drawing of the tilting mechanism at (top) neutral, (middle) tilt down, (bottom) tilt up positions; and (right) the centre distance between the V Groove Bearing and the guide rod.

The entire tilting and actuation module is built into a rigid aluminum frame to counteract torsional forces arising from the offset placement between the actuator support and the load points on the tabletop. This offset placement allows us to avoid stacking the load cells on top of the actuators, thus reducing the thickness of the mechanism and minimizing the visual impact of the mechanical parts on the overall impression. The structural elements are machined from high-strength 7075 aluminum alloy while the non-load-bearing components are 3D-printed in PLA. The bottom cover plate, which can be seen from the underside of the table, is made from 316 stainless steel and painted to match with the table top.

3 Motion Sensing and Control Algorithms

To control the tilt phase and direction in response to the glass object's motion, the system continuously estimates the object's position using four 100N load cells mounted beneath the tabletop at its corners (Fig. 6). This configuration, similar in principle to a pancake-style force-torque sensor used in robotic end effectors [Kim et al. 2017], measures the vertical force distribution. From these values, the system calculates the 2D centre of pressure (CoP) using the following weighted centroid formula:

$$x = \frac{\sum_{i=1}^N F_i x_i}{\sum_{i=1}^N F_i}, \quad y = \frac{\sum_{i=1}^N F_i y_i}{\sum_{i=1}^N F_i}$$

where F_i is the force measured at load cell i , and (x_i, y_i) are the known positions of the load cells in the table frame.

By sampling this position at 60Hz, the system constructs a real-time trajectory of the glass object. Numerical differentiation of the position signal yields estimates of angular velocity and acceleration. An exponential moving average filter is used for all the estimates to reduce noise. A digital dashboard (Fig. 7) was created for tuning the various filter parameters in the system, which is not shown to the viewers. The system also maintains a dynamic estimate of

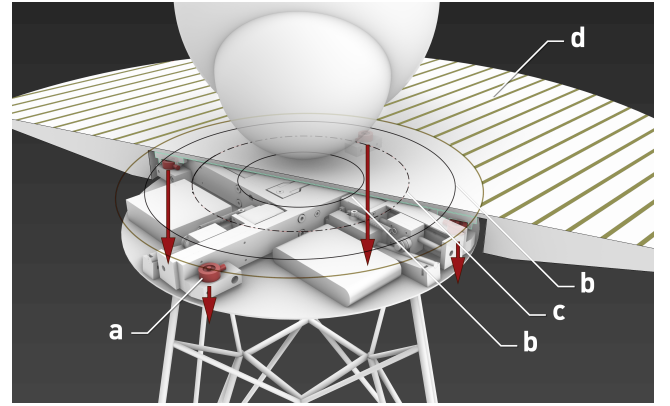


Figure 6: Diagram showing the arrangement of (a) four load cells at the corners of the mechanism, (b) contact tracks between the glass object and the table, (c) centre of pressure trajectory, and (d) keep out zone. Red arrows indicate the difference in forces experienced by the load cells.

oscillation period, used to filter motion reversals (velocity zero-crossings) and to classify motion states into three conditions:

- Stationary** – angular velocity remains below a threshold for several seconds.
- Oscillation** – characterized by motion reversals (detected by velocity changing sign) within a constrained angular range. Note that oscillation is caused by a small imbalance in the glass object and imperfection in its circular orbit. Thus creating a low potential energy point where the glass object naturally settles.
- Full Rotation** – detected when velocity retains consistent sign and position history accumulates past 2π . Typically, the angular velocity will be larger than $90^\circ/\text{sec}$.

Each regime triggers a different control policy. When stationary, the system gently tilts the table to initiate motion by creating a low point 90° ahead of the object. In the oscillation phase, energy is injected during the last 50% of the deceleration phase of the oscillation by creating a low point $20\text{--}30^\circ$ ahead of the object and reaching peak tilt as the object stops momentarily to reverse direction —much like pumping a swing at the right moment. In continuous rotation mode, tilt is maintained tangentially to the trajectory, reducing frictional loss while avoiding abrupt acceleration. Transitions between the three regimes follow a predetermined order from 1 to 2 to 3 as system energy is increased, and in reverse by simply stopping actuation. Alternatively, it is also possible to decelerate more aggressively by tilting the table behind the object, causing the object to lose energy rapidly.

The tilt vector is computed in XY coordinates (from the dynamically changing lead angle and tilting amplitude) and mapped to actuator positions through a forward kinematic model. In order to reduce computation time for a more responsive control loop, pre-computed look-up tables are used. The system remains within quiet velocity limits (under 200RPM motor speed), which corresponds to a tilt rate of approximately $1^\circ/\text{s}$ —imperceptible to viewers, but sufficient to accelerate and sustain the object's motion.

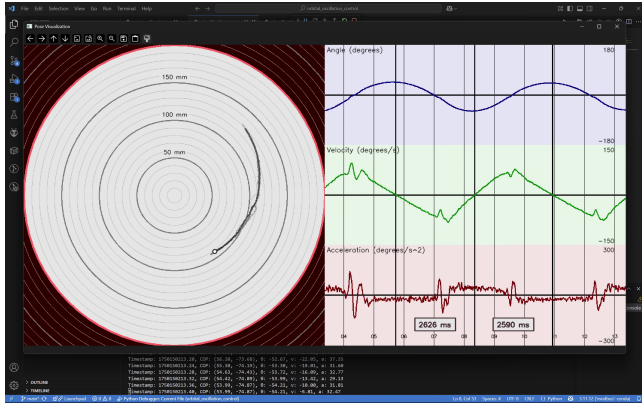


Figure 7: Dashboard of the sense and control system showing (left) visualization of the glass object's position and trajectory around the centre of the table relative to a safe zone and (right) time plot of the angular position, angular velocity, angular acceleration, oscillation reversal points, and their period.

The entire sensing and control loop is managed by a Raspberry Pi 4B (4GB), which samples the four load cells at 60Hz via a 24-bit analog-to-digital converter (ADC) using a Modbus RTU protocol. The ADC provides sufficient resolution and low latency for smooth state estimation. Communication with the closed-loop stepper controller is handled over a shared UART line with message acknowledgements turned off to reduce communication time. All filtering, derivative estimation, state classification, and tilt vector computation are executed on the control computer in real time. The system is designed for continuous operation and operates entirely untethered via two 24V battery packs (3400mAh each), providing approximately 8 hours of runtime when combined with occasional slow down and restarts. As a safety precaution, if the centre of pressure is further than 200mm away from the table centre, indicating a shift of the glass object away from the table centre, the tilting control would gradually slow down to a stop and wireless inform the operator to reposition the glass object.

4 Interaction Dynamics and Viewer Experience

From a viewer's perspective, Orbital Oscillation appears as a self-contained system—an enigmatic glass object quietly circling the centre of a minimalist table. The absence of visible motors, wires, or movement in the table itself heightens the sense of autonomy in the object's behavior. The tilt of the surface, limited to less than one degree in typical operation, ensures that the actuation remains invisible. This engineered invisibility fosters a perceptual ambiguity: viewers are drawn into the contradiction of seeing motion without seeing what causes it. This experience is carefully designed to oscillate between cognitive engagement and sensory surrender. At first, the sculpture appears passive—almost static. Then, slowly, the glass object begins to roll, completing partial arcs or gentle orbits. Viewers may notice the cyclical nature of the motion, forming predictions based on its rhythm. However, these predictions are subtly destabilized. Sometimes the object speeds up; sometimes it slows

or changes direction without warning. The resulting experience foregrounds the temporality of perception itself—what has just happened, what is happening now, and what might happen next. This dynamic aligns with the phenomenological framing of the work. Drawing from Edmund Husserl's triadic model of internal time consciousness—retention, impression, and protention—the sculpture unfolds as a layered perceptual event. Each motion becomes a site of negotiation between memory, presence, and expectation. The work generates assumptions of autonomous motion (e.g., battery-powered operation), later subverted by emergent evidence of external actuation. This phased revelation—from apparent independence to discovered relational movement—embodies a performative investigation of perception. By materializing this inference process through kinetic behavior, the sculpture renders perceptual scaffolding physically legible. The controlled variability introduced by the feedback system ensures that the viewer's cognitive model of the system is always slightly unstable, inviting closer attention and interpretive speculation.

5 Conclusion

Orbital Oscillation resonates with Jean Tinguely's *Méta-Harmonie* [Tinguely 1979] in its engagement with Husserl's triadic temporality. Both works exploit motion to stage a dialogue between predictability and disruption, inviting viewers to project rhythmic expectations onto mechanical behavior. Tinguely's complex assemblage of gears, motors, and wheels celebrates mechanized chaos, laying bare its causality to amplify absurdity and entropy. This was echoing the time it was created, where mechanisation came to an end, and the digital started. In contrast, Orbital Oscillation erases all traces of mechanisms. Its concealed actuators and silent control systems shift the phenomenology inward: retention holds only the glass object's prior arc; impression registers its current velocity; protention strains toward an anticipated orbit unhinged from visible causality. While *Méta-Harmonie* turns time into spectacle, Orbital Oscillation turns time into introspection.

Through the concealment of its control system, Orbital Oscillation creates an uncanny disjunction: a visible object in motion, with no apparent cause. This paradox draws the viewer into a temporal loop—not just of observation, but of internal synchronization. It demonstrates the complexity behind the perceived simplistic motions of a rolling glass object and how a tightly integrated mechatronic system establishes the continuation of a previously materialised structure. Once 3D printed. It is now dissolved as motion and into a medium for perceptual and philosophical reflection.

Rather than foregrounding its technical complexity, Orbital Oscillation channels engineering into a form of temporal poetics, constructing context without formalised physical structure as seen in the glass works before. Instead, subtle interventions shape expectation, and motion becomes a site of aesthetic and cognitive negotiation. It offers a model for a post-digital kinetic art, one in which artistic intention and mechatronic design are not layered, but co-extensive: bound together in the creation of experience itself.

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